

SANANDAJ, Iran, Islamic Republic of

WMO: 407470

Lat: 35.247N

Lon: 47.009E

Elev: 4522

StdP: 12.45

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.3	19.0	5.3	8.5	30.8	9.0	10.3	35.7	24.3	46.9	21.1	45.2	1.4	10	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.1	102.3	61.4	100.3	60.9	98.3	60.5	63.9	94.1	62.7	93.7	61.6	92.9	8.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.5	72.0	70.3	51.6	67.2	68.5	49.9	62.9	66.5	31.6	93.9	30.6	94.0	29.7	92.8	68.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	17.5	14.5	DB	8.8	105.9	6.0	1.4	4.5	106.9	1.0	107.7	-2.4	108.4	-6.7	109.4	(4)
(5)			WB	8.2	65.4	5.9	2.0	4.0	66.8	0.6	68.0	-2.7	69.1	-7.0	70.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.1	34.6	38.9	47.2	54.9	64.2	75.4	82.3	81.3	71.3	60.3	46.7	38.9		
	DBStd	17.22	7.46	6.89	6.41	5.67	4.74	4.70	3.89	3.86	4.58	5.17	6.00	5.78		
	HDD50	1414	477	313	129	19	0	0	0	0	0	1	129	346		
	HDD65	4121	942	730	552	304	72	0	0	0	5	159	548	809		
	CDD50	4381	0	3	43	167	440	762	1001	971	639	322	31	2		
	CDD65	1612	0	0	1	2	47	313	536	506	193	14	0	0		
	CDH74	26224	0	0	17	107	1069	5005	8278	7795	3387	565	1	0		
	CDH80	15063	0	0	0	20	311	2784	5239	4890	1703	116	0	0		
	WSAvg	4.8	4.5	4.9	6.2	5.6	5.0	5.3	5.2	4.8	4.2	4.1	3.5	3.7		
	PrecAvg	16.1	1.7	1.9	2.4	3.0	1.1	0.2	0.1	0.0	0.1	1.3	2.4	1.8		
(15) Precipitation	PrecMax	25.7	3.4	4.3	7.0	5.6	4.6	1.4	0.7	0.3	0.3	5.8	8.4	4.3		
	PrecMin	10.7	0.2	0.2	0.4	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1		
	PrecStd	3.5	0.8	0.9	1.3	1.5	1.0	0.2	0.2	0.0	0.1	1.3	1.7	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.3	62.8	76.7	82.7	89.8	100.8	105.5	104.1	97.2	86.3	71.4	62.9		
		MCWB	41.9	44.8	49.8	54.3	57.2	60.5	62.3	61.7	59.2	55.0	49.6	44.5		
	2%	DB	53.8	58.7	70.1	76.9	86.3	98.4	102.9	102.1	94.9	82.8	66.6	57.2		
		MCWB	41.0	43.0	47.7	53.2	55.9	59.8	61.6	61.3	58.1	53.8	48.2	43.3		
	5%	DB	50.3	55.1	66.0	73.2	83.8	95.3	100.8	100.2	92.3	80.4	63.0	53.4		
		MCWB	39.5	41.7	46.0	51.2	55.1	58.9	61.2	60.9	57.3	53.1	47.1	41.9		
	10%	DB	46.7	51.5	61.5	69.6	80.5	93.1	98.6	98.2	89.6	76.9	59.2	49.8		
		MCWB	38.4	40.4	44.4	50.2	54.8	58.1	61.0	60.4	56.6	52.1	46.0	40.6		
	0.4%	WB	45.4	47.8	51.3	56.6	61.3	64.2	65.9	65.6	61.4	57.1	53.1	47.6		
		MCDB	52.3	57.7	68.8	73.9	80.5	91.1	96.0	96.6	89.8	78.0	63.3	56.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.0	45.0	49.2	55.0	58.9	61.5	64.1	63.8	59.8	55.7	50.9	45.4		
		MCDB	50.0	54.6	66.1	71.2	77.7	91.1	94.6	94.7	89.4	75.3	59.8	53.6		
	5%	WB	41.1	43.0	47.4	53.4	57.4	60.2	63.0	62.5	58.4	54.5	49.3	43.3		
		MCDB	48.5	52.1	62.5	67.4	76.3	91.3	94.7	93.5	88.0	74.1	58.2	50.8		
	10%	WB	39.1	41.2	45.8	51.8	56.1	59.2	62.0	61.5	57.3	53.3	47.6	41.5		
		MCDB	46.3	49.6	59.2	65.5	75.0	90.1	93.8	93.2	86.4	72.8	57.1	48.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.7	21.6	23.7	24.9	29.2	34.5	33.1	34.7	36.6	31.0	23.4	20.5		
		MCDBR	25.7	27.9	31.8	32.1	35.3	36.7	36.1	36.6	38.7	36.8	30.7	26.9		
		MCWBR	14.5	14.4	14.1	12.5	11.5	11.1	10.7	10.9	13.1	14.4	14.7	14.8		
		MCDBR	21.8	23.0	26.7	26.0	28.9	33.6	31.6	32.4	35.9	30.7	23.1	22.4		
	5% WB	MCWBR	13.2	12.6	12.6	10.9	10.6	10.6	9.7	10.1	12.9	12.7	11.4	13.1		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.373	0.439	0.529	0.520	0.468	0.507	0.469	0.404	0.428	0.359	0.326		
	taud		2.246	2.099	1.939	1.765	1.811	1.943	1.853	1.937	2.094	2.006	2.200	2.274		
	Ebn at Noon		273	270	261	244	248	261	250	257	269	247	257	265		
	Edh at Noon		33	44	56	70	68	59	65	58	47	47	35	31		
(44) All-Sky Solar Radiation	RadAvg		877	1195	1524	1820	2212	2637	2484	2305	1969	1397	967	774		
	RadStd		69	108	123	150	150	116	83	78	87	106	116	89		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	+1.28	-2.81	-1.67	N/A	N/A	N/A	N/A

Nomenclature: See separate page